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Contents:
Basic counting principle, Rule of Sum, Rule of Product. Permutations, Combinations. Pigeon hole Principle, Binomial Theorem Recurrence rxn, generating functions.

The basic counting principles in discrete mathematics are foundational rules used to determine the total number of ways events can occur or tasks can be completed without listing every possibility.

eg: Consider guessing a 3-digit PIN. Each digit can be any number from 0 to 9, giving 10 possible choices for each digit. Thus the total number of possible PIN combinations is $10 \times 10 \times 10 = 1000$, which include all combinations from 000 to 999.

The Rule of Sum (Addition Principle)

→ The Rule of Sum states that if a task can be done in one of n ways or in one of m ways, and the two actions cannot be done at the same time, then there are n + m ways to choose one of the actions.

Keyword : "OR"
Condition : The choices must be mutually exclusive (disjoint)
Example : If a shelf has 5 math books & 3 science books, and you want to pick one book to read, you have 5 + 3 = 8 choices.

The Rule of Product (Multiplication Principle)

→ The Rule of product states that if a procedure consists of two sequential tasks, where the first task has n ways and the second task has m ways, then there are n × m ways to perform

Keyword : "AND"
Condition : The tasks are performed sequentially or independently.
Example : If you have 3 shirts and 4 pants, an outfit combination of one shirt and one pair of pants gives $3 \times 4 = 12$ total outfits.

Factorial
- factorial is the product of an integer and all the positive whole numbers below it, down to one.
- Symbol : Denoted by an exclamation mark (!) placed after a number.

Formula : $n! = n \times (n-1) \times (n-2) \times \dots \times 1$
Examples
$3! = 3 \times 2 \times 1 = 6$
$4! = 4 \times 3 \times 2 \times 1 = 24$
$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

DATE

Permutation

- Permutation determines the number of possible arrangements for a specific set of elements.
- In permutation, order matters. for example (2,1) and (1,2) are counted as different.

Example

I have 3 objects with me:

A B C

from these 3 objects, I have to ~~ARRAG~~ ARRANGE 2 objects.

So the arrangements are: AB BA
BC CB
AC CA

here we can arrange in 6 ways

$$^{3}P_{2} = \frac{3!}{(3-2)!} = \frac{3!}{1!} = \frac{3 \times 2 \times 1}{1} = 6$$

The permutation formula is used to calculate the number of ways to arrange a subset of objects from a larger set where the order of selection matters.

$$^{n}P_{r} = \frac{n!}{(n-r)!}$$

where,

n = number of total objects
r = number of objects chosen at once

Combination

- A combination is a selection of items from a larger group where the order of selection does not matter.
- In combination order does not matter.

picking items like A and B is the same as picking B and A.

Example

I have 3 objects with me:

A B C

out of these 3 objects, I have to SELECT 2 objects.

So the selections made are } = AB, BC, AC

here we can select in 3 ways

$$^3C_2 = \frac{3!}{2!(3-2)!} = \frac{3!}{2!\,1!} = \frac{3\times2\times1}{2\times1} = 3$$

The combination formula is used to choose 'r' components out of a total number of 'n' components and is given by:

$$^nC_r = \frac{n!}{r!\,(n-r)!}$$

where,

n = number of total objects

r = number of objects you want to select.

DATE

Pigeonhole Principle

The pigeonhole principle states that if n items are put into m containers, and $n > m$, then at least one container must contain more than one item.

Simple idea: If you have 5 pigeons and 4 pigeonholes, putting every pigeon in a hole means at least one hole gets two pigeons.